DETERMINING SONIC ACTIVITY IN ELECTROACOUTIC MUSIC

David Hirst
School of Contemporary Music
University of Melbourne
dhirst@unimelb.edu.au

ABSTRACT

This paper seeks to extend the author's SIAM Framework for the analysis of electroacoustic music by posing the following question: What propels an electroacoustic work along from moment to moment, section to section, or scene to scene, given the absence of harmonic cadences associated with traditional tonal music? Taking as its subject matter, Jonty Harrison's piece *Unsound Objects,* the paper provides a brief outline of the SIAM Analysis Framework, describes a previous analysis of *Unsound Objects* using this method, and then extends the analysis to seek a measure for "sonic activity" as an indicator of periods of tension and release within the work. The aim of the current study is to seek an alternative method to the use of "spectral irregularity" for measuring sonic activity in electroacoustic music. This paper combines a number of time-related tools from the program Sonic Visualiser to create Inter-onset Time and Inter-onset Rate plots for *Unsound Objects.* The paper concludes that determining inter-onset time can give us a plot that is useful in showing the main sections within a work. Calculating its reciprocal, inter-onset rate, can generate a graph that provides some measure of the varying activity within an electroacoustic work. The next step in research is to seek to display further subtleties in the organisation of time, eg. Rhythmic patterns.

1. DEFINING THE PROBLEM

While undertaking a recent analysis of Jonty Harrison's electroacoustic musical work, *Unsound Objects* (Hirst 2013) the initial phase involved analysing the acoustic surface to identify sound objects. The next phase required an examination of relationships between sound objects, giving rise to the following question: What propels the work along from moment to moment, section to section, scene to scene? To help answer this question, I observed that an increase in sonic activity seems to elicit expectation in the listener that an important event is about to occur. There is a build up that seems to require a release the longer the build up goes on. But how can we measure something I have called "sonic activity" and, even better, how can we display sonic activity easily within a work? Can some form of signal processing be used and be represented to assist in the interpretation of electroacoustic musical works?

In the analysis of the Harrison piece, I began to address these questions, but there was not sufficient space within the confines of that article to discuss the issues in any great depth. This paper provides an opportunity to take the discussion further, and thereby expand and refine the author's SIAM framework for the analysis of electro-acoustic music. In the following sections, I will briefly describe Harrison's *Unsound Objects,* recap on the definition of the SIAM (Segregation, Integration, Assimilation and Meaning) framework, and then elaborate on what I call horizontal integration, which involves an examination of the temporal dimension of an electroacoustic musical work. This will lead on to a discussion of sonic activity and some possible ways of representing it.

2. HARRISON'S *UNSOUND OBJECTS*

Unsound Objects is a substantial work of some 12 minutes 59 seconds (779 secs). First performed as a commission for the 1995 International Computer Music Conference in Banff, the version used in this analysis appears on the Articles indéfinis CD (Harrison 1996). While often regarded as a traditional "acousmatic" composer, in a 2002 interview Harrison said this about the use of recognisable, recorded sounds in *Unsound Objects:* 'As it would therefore be impossible to expect listeners to be able to exercise "reduced listening" in such a framework I chose instead to exploit this very recognisability in working on the piece. This is an implicit challenge to Schaefferian orthodoxy about the definition of the sound object through reduced listening; it is dangerous and, hence, "unsound".' (Palmer 2002).

An analysis of the whole of *Unsound Objects* was carried out using a modified form of the author's cognition-based SIAM framework for the analysis of electroacoustic music. The detailed analysis of *Unsound Objects* is available in that previous article (Hirst 2013), but here I will briefly recap some of the SIAM concepts in order to provide some sort of context.

3. THE SIAM ANALYSIS FRAMEWORK

The SIAM framework (Hirst 2008) stands for Segregation, Integration, Assimilation and Meaning. It can be briefly summarised as follows:

Segregation: identification of sound events and the factors responsible for identification.

Integration – Vertical: consider vertical integration and segregation as a cause of timbre and texture variance; consider psychoacoustic dissonance and musical dissonance; consider emergent properties relating to pitch (horizontal overlap).

Integration – Horizontal: identify sequential streams and patterns of sonic objects; determine causal linkages, relationships and possible syntaxes; consider organisation in time and the horizontal integration of pitch (where appropriate).

Assimilation and Meaning: consider various types of discourse, e.g. discourse on the source-cause dominant (semantic) to typological-relational dominant (syntactic) continuum; consider global organisation in time and any hierarchical relationships; consider expectation-interruption, arousal and meaning, semiotic relations, etc.

4. ANALYSIS OF *UNSOUND OBJECTS*

Using the SIAM framework as a starting point, the analysis of *Unsound Objects* was adapted according to the directions dictated by the work itself.

The first part of the analysis can be described as **analysing the acoustic surface**. This involved "segmentation". Large scale segmentation into sections, and then small-scale segmentation of sound events from each other. The spectrogram and audio waveform displays were useful for the process. Sound events were annotated on the spectrogram and it was possible to get a time-stamped listing of the annotation layer, using the program Sonic Visualiser (Cannam et al 2010), which was then imported into a spreadsheet program (Microsoft Excel) and printed as a listing of all the annotations. The visual screens and printed time-stamped sound object listings became the data that facilitated detailed identification and specification of sound events within the aurally identified sections of the work.

The next phase of the analysis involved **moving beyond the acoustic surface** to examine structures, functions and motions between sound events. By "zooming out" to look at longer sections of the work, or carrying out "time-span reduction", we can observe changing sonic patterns over the course of the work. We can look at the different sections and ask questions like: What propels the work along from moment to moment, section to section, or scene to scene? To help answer this question, we can observe that an increase in sonic activity seems to elicit expectation in the listener that an important event is about to occur. But how can we measure and, even better, display activity within a work? Well the Sonic Visualiser program provides access to a suite of plugins of signal analysis. In the *Unsound Objects* article, I postulated that the type of analysis that seems to correlate best with sound object activity is a plot of "spectral irregularity" versus time.

There are several different methods for calculating the irregularity present within a spectrum, but essentially they both give a measure of the degree of variation of the successive peaks of the spectrum. Jensen, for example, calculates the sum of the square of the difference in amplitude between adjoining partials (Jensen 1999). What I am postulating here is that where there is a large variation across the spectrum, partial to partial, then this can provide us with a depiction of a high degree of activity. Figure 1 depicts an irregularity plot for the whole of *Unsound Objects*. Within that article I noted that it could be argued that we could simply use a plot of overall amplitude, or one of its variants, but I asserted that simply using amplitude tends to average out "activity", whereas spectral irregularity provides a much more fine-grained representation of activity from instant to instant through time. Subsequent work showed that RMS Amplitude plots give very similar graphs to those of irregularity. I shall return to this point later on.

The analysis of *Unsound Objects* then combined the use of spectral irregularity plots with aurally identified sections, within the work, to provide a detailed analysis of "activity" and to tabulate "sound types" for each section. This table showed "activity amount and type" and "selected sound object types".

The work actually divides into two main halves and after the two halves were compared, a summary of sonic archetypes (in the form of mimetic archetypes and structural archetypes), sound transformations, functional relations, and sonic activity was discussed.

In the final discussion section of the *Unsound Objects* analysis article, I made a number of observations and conclusions, and then ended with this: "Finally, we have used spectral irregularity plots to assist with the identification and isolation of periods of activity within different sections of the work. It seems as though it may be a useful measure for the analysis of other works – a hypothesis remaining to be tested."

This current article documents the follow up work to examine how best to represent and analyse "activity" within an electroacoustic music work.

5. ALTERNATIVE METHODS FOR DETERMINING ACTIVITY

The aim of the current study is to seek an alternative method to the use of "spectral irregularity" for measuring activity in electroacoustic music.

In essence, activity could be defined as the number of sound events in a given time period. Therefore we are interested in the onset time of each sound event, and its duration. Let's start with onset time. What signal analysis tools exist for determining sound event onset time within a musical work?

The program Sonic Visualiser, has a number of tools within it to perform such an analysis. Aubio onset detection (aubio.org) has eight different types which all

produce a single list of time "instants" (vertical lines when plotted) of individual start times. This output can be exported to a spreadsheet. Their algorithm can be varied to suit the source material. The Queen Mary, University of London, in-built Sonic Visualiser onset detection algorithm lists three types of onset detector, but these are just the one detector with lots of variables: Program; Onset detection function type; Onset detection sensitivity; Adaptive whitening; Channel options for stereo files; Window size; Window increment; and Window shape. Output is an "onset detection function" which is a probability function of a "note" onset likelihood.

In developing a method for the detection of onsets in *Unsound Objects,* combining several forms of representation was found to provide a more reliable guide to data gathering rather than using any single plot. After some experimentation, the following combination was employed, using the Queen Mary algorithms:

1. RMS Amplitude (or Spectral Irregularity).
2. Smoothed detection function: Time Values (displays probability function of onsets).
3. Note onsets: Time Instants. Program: Soft Onsets; Onset detection function: Complex Domain; Onset detection sensitivity: 60%; Adaptive whitening: Yes.

This resulted in the onsets (#3 above) aligning pretty well with the smoothed detection probability (#2 above), but with some low level noise swells failing to trigger the onset detector (#3 above).

The "time instants" data (#3 above) was exported, then imported into an Excel spreadsheet in order to be able to make further calculations such as "inter-onset times" (the time between onsets). Figure 2 shows a plot of Inter-onset Time versus Time for the whole of *Unsound Objects.* Its peaks show us where there are long breaks in the work, and give a pointer to how the work may be divided up in analysis.

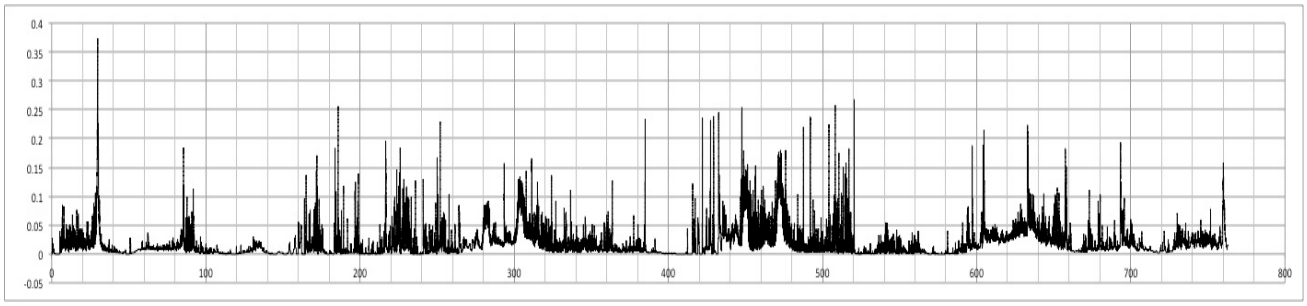


Figure 1. A spectral irregularity plot for the whole of *Unsound Objects*.

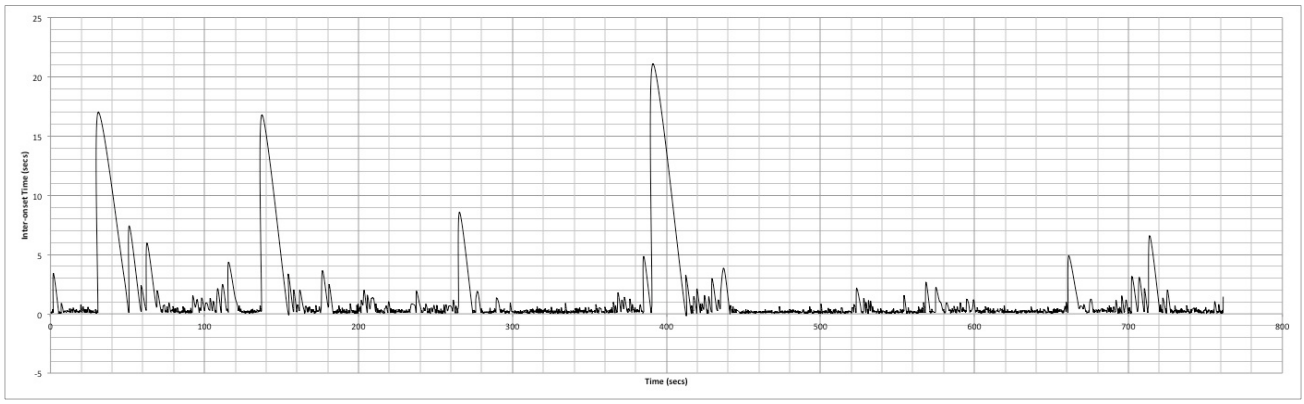


Figure 2. Plot of Inter-onset Time vs Time (secs) for the whole of of *Unsound Objects*.

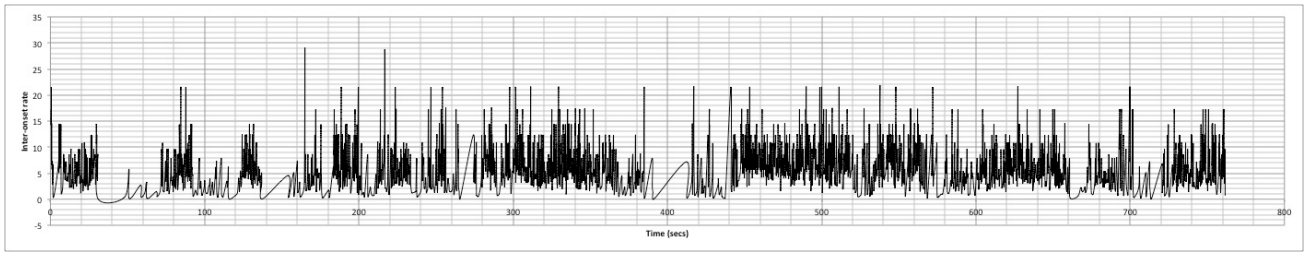


Figure 3. Plot of Inter-onset Rate vs Time (secs) for the whole of *Unsound Objects*.

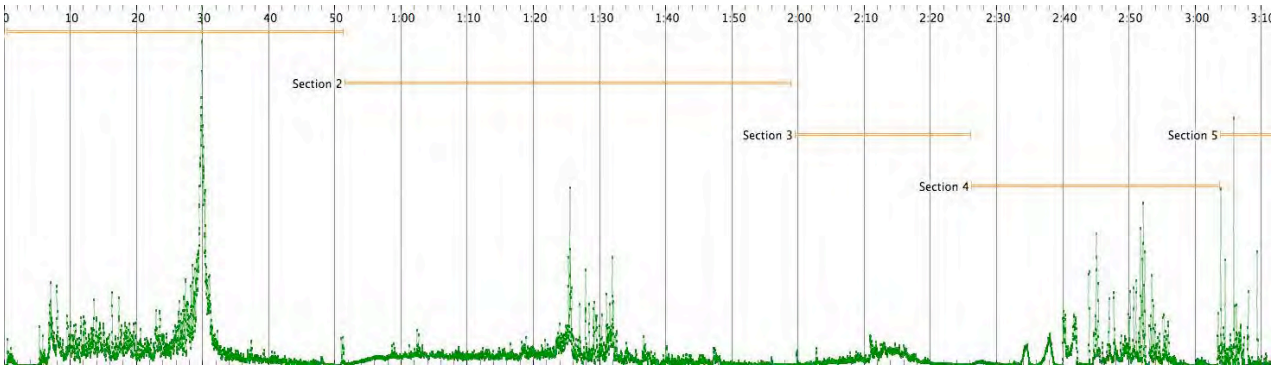


Figure 4. An irregularity plot for the first four sections of *Unsound Objects*. (Note: Time scale is shown in min:sec)

Displaying time instants, however, only progresses us part of the way to obtaining a measure of event "activity". Inter-onset "rate" was then calculated and plotted, as shown in Figure 3. This provides us with a measure of the number of onsets per second, which, in turn, provides a guide to the amount of event initiation activity at a particular time within the work.

6. OBSERVATIONS AND CONCLUSIONS

Determining inter-onset time can give us a plot (Figure 2) that is useful in showing the main sections within a work. Calculating its reciprocal, inter-onset rate can generate a graph that provides some measure of the varying activity within an electroacoustic work (Figure 3). If we had graphed Figure 3 at the beginning of the analysis, we would have observed that the piece does divide into two, with little activity between about 390 and 410 seconds. The first half begins with three bursts of activity, followed by a longer, more active phase of increasing activity until the "mid-break". The second half is more continuously active until around 660 seconds, where the work has several less active periods, perhaps in preparation for the end of the piece.

In the previous analysis of *Unsound Objects,* sections were first determined aurally, then superimposed over the irregularity plot (for an example this for the first four sections of the piece, see Figure 4). Comparing the plot of inter-onset rate (Figure 3) with the irregularity plot for the whole of the work (Figure 1), we can see that the piece appears to be much more active in figure 3 than figure 1, especially in the second half. The question remains as to which is a better measure of "activity"? The inter-onset rate is probably a more accurate method, but it seems exaggerated. This is possibly because it doesn't take into account the loudness of the events. Perhaps if this plot (Figure 3) was modified by the RMS amplitude, then a more useful picture of "effective activity" may emerge. There are also inherent definition problems for "iterative" sound events, such as drum rolls or machine sounds. Is such a sound type one long event or many short events? This phenomenon may skew the events per second data.

In terms of automating analysis, the inter-onset time plot (Figure 2) is very effective in identifying sections in a long musical piece, while the inter-onset rate (Figure 3) does provide a measure of active versus inactive depiction for various passages in a long piece.

The next step in this work is to extend temporal analysis further by looking at finer details within sections (see Hirst 2014).

7. REFERENCES

Cannam, C. Landone, C. & Sandler, M. 2010. "Sonic Visualiser: An Open Source Application for Viewing, Analysing, and Annotating Music Audio Files." *Proceedings of the ACM Multimedia 2010 International Conference*.

Harrison, J. 1996. *Unsound Objects*. On *Articles indéfinis*. IMED 9627. emprintes DIGITALes CD.

Hirst, D. 2008. *A Cognitive Framework for the Analysis of Acousmatic Music: Analysing Wind Chimes by Denis Smalley* VDM Verlag Dr. Muller Aktiengesellschaft & Co. KG. Saarbrücken.

Hirst, D. 2013. "Connecting the Objects in Jonty Harrison's Unsound Objects*." eOREMA Journal, Vol 1*. http://www.orema.dmu.ac.uk/?q=eorema_journal.

Hirst, D. 2014. "The Use of Rhythmograms in the Analysis of Electro-acoustic Music, with Application to Normandeau's Onomatopoeias Cycle". Paper submitted to ICMC 2014.

Jensen, K. 1999. *Timbre Models of Musical Sounds*, Ph.D. dissertation, University of Copenhagen, Rapport Nr. 99/7.

Palmer, J. 2002. In Conversation with Jonty Harrison. *21st Century Music*, Vol. 9/1 (USA, January 2002). Republished in eContact! 10.2 Entrevues Interviews(1). Montréal: Communauté électroacoustique canadienne Canadian Electroacoustic Community, August 2008. http://cec.sonus.ca/econtact/10_2/index html